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Research on Prediction Method of Cable Tower Pre-bias Based on Model Fusion

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Abstract

Traditional construction monitoring is labor-intensive and faces the problems of lengthy calculation cycles and high experience requirements. Oriented by data-driven, a method for pre-bias prediction of cable towers based on model fusion is proposed. First, collect and build an experimental data set containing 125 samples. After data preprocessing, establish SVR and KNN models with initial parameters. And conduct model evaluation after cross-validation and hyperparameter optimization. Second, utilize the mean absolute error (MAE) to define the loss function of the fusion model, and obtain the fusion weight of the weak learner. The weighted average method is used to fuse the SVR and KNN models. The deep integration of SVR and KNN is realized at the model decision level, which further improves the model fitting performance and the reliability of the prediction results. Finally, the performance of each model is evaluated based on the proposed comprehensive evaluation index. The experimental results demonstrate that the SVR and KNN models perform well on tasks with small sample sizes, and the fusion model has an MAE of 1.361, $MAPE$ of 0.149, $RMSE$ of 2.08, and R^2 of 0.997 on the test set. The research method and content have greatly improved the precision and efficiency of the pre-bias control of the tower. Which can be used to improve and extend the existing finite element analysis method.

Keywords：Construction monitoring, tower pre-bias, model fusion, model evaluation, machine learning

1. Introduction

The role of the cable tower in the long-span bridge is crucial, and its stress situation is directly related to the safety and stability of the overall structure of the bridge. As an important indicator for the monitoring of the force and system of the tower, the alignment of the building is not only affected by many factors such as material properties and natural environment but also with the increase in the construction height of the tower, the subsequent construction segments of the building will also become unstable. To avoid the influence of horizontal displacement on the current construction section, a specific pre-bias needs to be set during the construction of the cable tower to ensure that the linear shape of the cable tower meets the design standards and specification requirements after completion.

In traditional construction monitoring, the control of the pre-bias of the tower is mainly based on finite element analysis. The calculation process of this method is cumbersome and time-consuming, and it also puts forward high requirements for the monitoring personnel's calculation ability and engineering experience[1]. Today, where the digital construction industry has an unprecedented layout, artificial intelligence technology has also brought new historical opportunities for the development of civil engineering[2-3] and monitoring of industrial processes[4]. Wang Jingchun et al.[5] used the Support Vector Regression (SVR) algorithm, the decision tree algorithm and the Extreme Gradient Boosting (XGBoost) algorithm to predict the pre-chamber of continuous rigid frame bridges and used the mean absolute Based on the evaluation indicators such as error (MAE), root mean square error ($RMSE$) and coefficient of determination (R^2), the model performance was

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compared and studied; Liu Yang et al.[6] analyzed the reliability of the bridge anti-vibration force based on the SVR algorithm, which improved the bridge's anti-vibration resistance. The efficiency of dynamic reliability analysis and calculation; Xu Xuesong et al.[7] predicted the pre-camber of long-span continuous girder bridges based on the BP neural network, which improved the linear control ability of long-span bridges; Wang et al.[8] based on MEC-BP The neural network is used to predict and analyze the deflection of the corrugated steel web composite box girder bridge. The experimental results show that the prediction accuracy of this method is higher, and the efficiency is much higher than that of the finite element analysis method; Wang et al.[9] The time-varying fatigue reliability of welded joints was evaluated, the influence of traffic flow and natural corrosion on the time-varying fatigue reliability of welded joints was deeply studied, and the efficiency and feasibility of the data-driven method were proved. The Bayesian dynamic linear model is used to model and predict the stress and strain of large-span bridges[10], and good prediction performance is obtained.

Machine learning methods have been successfully applied in bridge engineering construction monitoring. However, because engineering big data construction and data-driven management and decision-making research are still in their infancy, there are still some problems in existing research: (1) It is difficult and costly to obtain construction monitoring data. Using complex models such as XGBoost, BP neural network, and ANN on such tasks with few samples is prone to overfitting. However, solving this problem through an intelligent optimization algorithm (for example, MEC) will undoubtedly lead to an increase in computing resources and time costs; (2) The comparative study of multiple models only realizes the measurement of the performance differences of each model, but cannot To ensure the reliability of model prediction results; (3) The applicability of traditional regression model evaluation indicators and their characteristics are different, and comprehensive evaluation of different models cannot be achieved.

Therefore, in this study, based on comprehensively considering the influencing factors such as material properties and natural environment, based on the SVR and K-Nearest Neighbor (KNN) regression algorithm, a pre-bias prediction model of the cable tower was established. Then, multi-model fusion is performed based on the weighted average method, thereby improving the model performance and the reliability of the prediction results. Finally, a comprehensive multi-index model evaluation method is used for model evaluation. The experimental results show that the fusion model's prediction accuracy and generalization performance has been further improved, and related research can provide new ideas and solutions for the construction monitoring of large-span bridges.

2. Dataset preparation

2.1 Engineering background

This research conducts an empirical study based on the Xian Xin Road Crossing the River Channel Project, located between the Nanjing Second Yangtze River Bridge and the Fourth Yangtze River Bridge, with a total length of about 13.17 kilometers. The main bridge of the project is a double-tower single-span steel box girder suspension bridge with a main span of 1760 m, and the cable tower is a portal structure. The south and north towers adopt portal frames, group pile foundation, the diameter of a single pile is ϕ2.8 m, and there are 66 in total. Among them, the design pile length of the south tower is 101 m, and the design pile length of the north tower is 110 m. The central tower is made of concrete, and the upper beam is N-shaped. The tower height is 208.6 m above the bridge deck and 55.2 m below the bridge deck. The schematic diagram of the entire bridge structure is shown in Figure. 1.

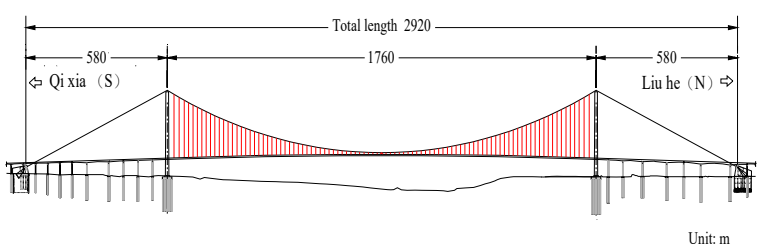

Figure 1 Main structure of Xian Xin Road Yangtze River Bridge

2.2 Dataset description

The experimental data set used in this study is extracted from the finite element analysis model of the Xianxin Road Yangtze River Bridge. Combined with the on-site construction progress and finite element analysis results, 125 sets of cable tower pre-bias data have been collected. And each collection of data included nine input features and one output

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variable. Among them, the input features are composed of building material properties and environmental factors, namely beam length (L), beam mass (G), web thickness (F), base plate thickness (D), concrete bulk density (γ), Modulus of elasticity (E), compressive strength (P), and temperature (T) and wind load (W); output variable is pylon pre-bias (CL). The experimental data's numerical statistics and correlation analysis results are shown in Table 1 and Figure. 2.It can be seen from the statistical analysis results in Table 1 that the minimum mass of the beam segment is about 411 kN , the maximum is about 7678 kN , and the standard deviation is 1435; at the same time, it can also be found that for different combinations of input variables, the pre-bias (CL) has an extensive variation range with a minimum value of 0.029 mm and a maximum value of 103.574 mm.

Table 1 Statistic features of experimental dataset

Direction	Description	Symbol	Unit	Min	Max	Mean	STD
Input	beam length	L	m	0.250	6.000	2.145	1.015
	beam quality	G	kN	411.428	7678.020	2997.116	1435.100
	web thickness	F	mm	120.000	250.000	172.902	44.739
	plate thickness	D	mm	120.000	250.000	172.902	44.739
	bulk density	γ	kN/m^3	23.354	26.266	25.101	1.257
	elastic Modulus	E	N/mm^3	3571.713	3623.963	3602.820	23.994
	compress-strength	P	N/mm^2	36.678	39.902	38.612	1.392
	temperature	T	℃	19.380	20.000	19.596	0.157
	wind-load	W	N/mm	5.624	56.578	36.798	14.339
Output	pre-bias	CL	mm	0.029	103.574	57.347	38.105

The correlation analysis results in Figure. 2 also show that some feature variables in the current dataset have high linear correlations. For example, the correlation coefficient of beam length (L) and beam quality (G) is 0.89, while the concrete bulk density (γ), elastic modulus (E) and compressive strength (P) also showed a high degree of linear correlation, and the Pearson correlation coefficient reached 1. Similarly, wind load (W) and temperature (T) also achieve strong correlations with pre-bias (CL) with an absolute value of 0.98. The linear correlations between the remaining feature variables and the pre-bias (CL) are generally low, indicating that the relationship between the cable pre-bias and these input variables is complex. At the same time, it also indirectly shows that the pre-bias prediction result of the tower based on a single model decision may not meet the requirements of high reliability of construction control.

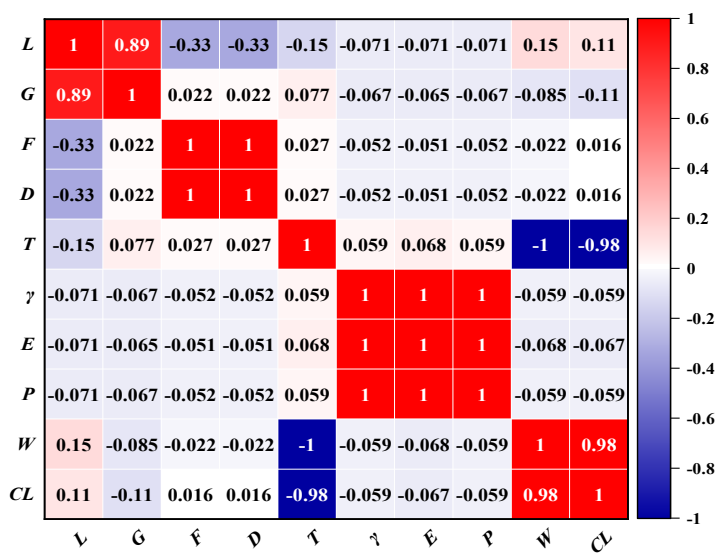

Figure 2 Correlation coefficient of the dataset

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2.3 Dataset preprocessing

To ensure the data integrity, the missing samples and duplicate samples in the dataset were first filtered. Secondly, the outliers in the data set are identified and processed using the boxplot visualization method. In addition, it can be seen from Table 1 that there are significant numerical differences in the extreme values of some input features. To eliminate the dimensional influence between the features and avoid data leakage, this study splits the data at a ratio of 7:3, using the Min-Max-Scaler method scales the training set data in equal proportions, and the calculation process is shown in formula.1.

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{1}$$

where x_{min} and x_{max} is the minimum and maximum values of the original feature, respectively. Taking the beam quality (G) as an example, the data distribution before and after normalization is shown in Figure 3. After the normalization process, the beam segment quality (G) value is more uniform in scale, and no negative values were introduced into the transformed data.

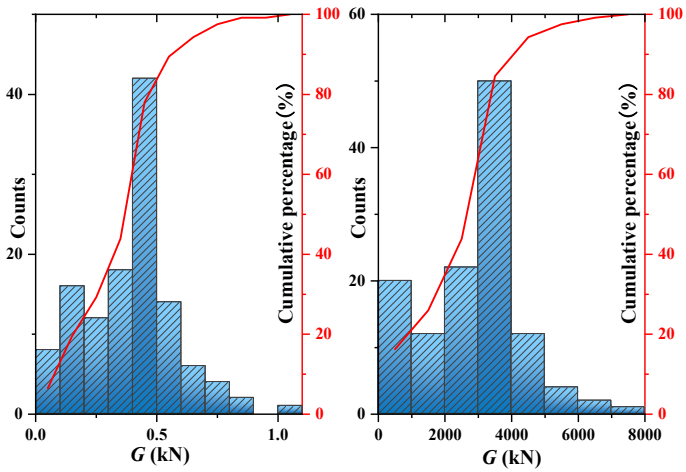

Figure 3 The data distribution of beam quality (G)

3. Methodology

Figure 4 shows the cable tower pre-bias prediction's overall modeling process based on this study's model fusion method. First, the SVR and KNN regression models with initialization parameters are constructed based on the training set data, and the model is evaluated on the test set after 5-fold cross-validation; secondly, the model hyperparameters are adjusted using Bayesian optimization until the model converges; then the optimized model is fused by the weighted average method.

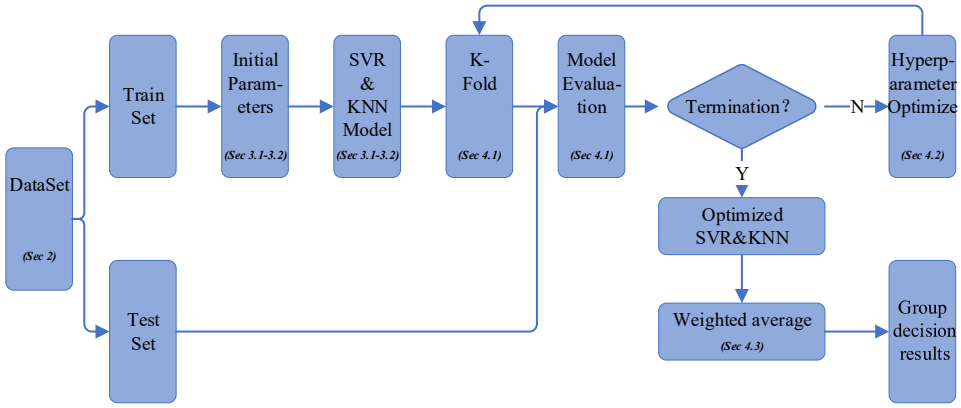

Figure 4 Modeling analysis flow chart

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3.1 Support Vector Regression

The SVR algorithm can handle small sample data well and has a strong generalization ability[11].。For given dataset $D=\{(x_i,y_i)|\ i\in\{1,2,\cdots,n\}\}$, $x_i\in R^n$, $y_i\in R$, The goal is to find a linear regression equation $f(x_i)$ that can adequately fit all samples, as in Equation 2, while minimizing the overall variance of all samples in the regression hyperplane.

$$f(x_i)=\omega\cdot\phi(x_i)+b \tag{2}$$

where ω is the weight vector, $\phi(x_i)$ is the nonlinear mapping function, and b is the bias. Combined with structural risk minimization, Equation 2 is equivalent to solving the optimization problem shown in formula. 3.

$$\frac{1}{2}\|\omega\|^2+C\sum_i^n L\left(f(x_i),y_i\right) \tag{3}$$

where L is the loss function, C is the penalty factor, which is used to impose penalties on the model to control the model complexity. SVR also improves the tolerance of the model by introducing slack variables $\{\xi_i\}_{i=1}^n$ and $\left\{\xi_i^*\right\}_{i=1}^n$, and further, a loss function in the form of Equation 4 can be obtained as follows:

$$\begin{gathered}\min R(\omega,b,\xi)=\frac{1}{2}\|\omega\|^2+C\sum_i^n\left(\xi_i+\xi_i^*\right)\\ \text{s.t.}\begin{cases}y_i-\omega\phi(x_i)-b\leqslant\varepsilon+\xi_i\\ \omega\phi(x_i)+b-y_i\leqslant\varepsilon+\xi_i^*\\ \xi_i,\xi_i^*\geqslant 0\end{cases}\end{gathered} \tag{4}$$

where ε is the insensitivity loss factor Using the Lagrange multiplier method, the dual form of the above equation is obtained as follows:

$$\begin{gathered}\max R\left(\alpha_i^*,\alpha_i\right)=-\frac{1}{2}\sum_{i,j}^n\left(\alpha_i^*-\alpha_i\right)\left(\alpha_j^*-\alpha_j\right)\phi(x_i)\phi(x_j)-\\ \sum_i^n\alpha_i(y_i+\varepsilon)+\sum_i^n\alpha_i^*(y_i-\varepsilon)\\ \text{s.t.}\begin{cases}\sum_{i=1}^n\left(\alpha_i-\alpha_i^*\right)=0\\ 0\leqslant\alpha_i,\alpha_i^*\leqslant C\end{cases}\end{gathered} \tag{5}$$

where α_i^* and α_i are Lagrange multipliers. The above convex quadratic programming problem is obtained by Mercer's theorem and the SVR expression is obtained, as shown in formula 6 , where $K(x_i,x)$ is kernel function.

$$f(x)=\omega\phi(x_i)+b=\sum_i^n\left(\alpha_i-\alpha_i^*\right)K(x_i,x) \tag{6}$$

3.2 K nearest neighbor regression

KNN is an instance-based supervised learning algorithm[12] suitable for classification and regression tasks. Since it does not need to learn a specific discriminant function from the training data, KNN is sometimes called a "lazy learner"[13]. Based on this feature, KNN does not have high requirements on the amount of data. As shown in Table 2, the input of the KNN algorithm is a data set with labels, and the algorithm's output is the sample's prediction result. The output result can be a discrete category value or a continuous prediction value depending on the specific task type. The main steps include three elements: hyperparameter K value, distance metrics, and model decision criteria.

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Table 2 Pseudo code of KNN algorithm

Input：Labeled dataset D
1. Choose hyperparameter K and distance metric； 2. Find the K samples closest to the sample to be predicted； 3. Obtain prediction results by majority voting method or simple average method；
Output：The class or prediction of the sample

Where the commonly used method for selecting the K value is the elbow method or cross-validation; the determination of the model decision criterion depends on the specific task. For classification tasks, the majority voting method is generally used; for regression tasks, the average method is usually used. The distance measurement is often used Euclidean distance. Let the Minkowski distance between two n-dimensional vectors $\mathfrak{X}$ and $\mathfrak{Y}$ is Dis, its definition is shown in Equation 7, and the Euclidean distance is a particular case of $p=2$.

$$Dis(\mathfrak{X},\mathfrak{Y})=\sqrt[p]{\left(\left|\mathfrak{X}_1-\mathfrak{Y}_1\right|\right)^p+\left(\left|\mathfrak{X}_2-\mathfrak{Y}_2\right|\right)^p+\ldots+\left(\left|\mathfrak{X}_n-\mathfrak{Y}_n\right|\right)^p}=\sqrt[p]{\sum_{i=1}^{n}\left(\left|\mathfrak{X}_i-\mathfrak{Y}_i\right|\right)^p} \tag{7}$$

3.3 Model Fusion

Model fusion is constructing and integrating several weak learners to achieve joint decision-making based on a specific fusion strategy and is sometimes referred to as a multi-classifier system[14]. As shown in formula 8, the average method is the commonly used fusion strategy in regression problems; that is, the fusion model $H(x)$ is the average of the prediction results of the base model $h(x)$.

$$H(x)=\frac{1}{T}\sum_{i=1}^{T}w_i h_i(x) \tag{8}$$

Where T is the number of base models, and w_i ($w_i \geq 0$) is the weight of the base model $h_i(x)$. In practice, $\sum_{i=1}^{T}w_i=1$ is usually set. When $w_i=\frac{1}{T}$, it indicates that the importance of the base model is the same or the performance is close; and when the base model's performance is quite different, more considerable weight is generally given to the model with stronger performance[15].

3.4 Performance evaluation

3.4.1 Conventional method

Mean absolute error (MAE), mean absolute percent error ($MAPE$), root mean square error ($RMSE$), and coefficient of determination (R^2)[16] are the four most frequently used evaluation indicators for regression models. as follows.

$$MAE=\frac{1}{S}\sum_{i=1}^{S}\left|Y_{i_{\text{observed}}}-Y_{i_{\text{predicted}}}\right| \tag{9}$$

$$MAPE=\frac{100\%}{S}\sum_{i=1}^{S}\left|\frac{Y_{i_{\text{observed}}}-Y_{i_{\text{predicted}}}}{Y_{i_{\text{observed}}}}\right| \tag{10}$$

$$RMSE=\sqrt{\frac{1}{S}\sum_{i=1}^{S}\left[\left(Y_{i_{\text{observed}}}-Y_{i_{\text{predicted}}}\right)\right]^2} \tag{11}$$

$$R^2=1-\frac{\sum_{i=1}^{S}\left(Y_{i_{\text{predicted}}}-Y_{i_{\text{observed}}}\right)^2}{\sum_{i=1}^{S}\left(Y_{i_{\text{observed}}}-\bar{Y}_{\text{observed}}\right)^2} \tag{12}$$

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where S is the sample size, that is, the total number of experimental data; $Y_{i_{\text{observed}}}$ and $Y_{i_{\text{predicted}}}$ represent the actual observed value and model prediction result respectively, namely the actual pre-bias of the tower and the pre-bias predicted by the model; $\overline{Y}_{\text{observed}}$ is the mean value of the actual pre-bias.

3.4.2 Comprehensive evaluation method

According to Section 3.4.1, when MAE, $MAPE$ and $RMSE$ are closer to 0, and the coefficient of determination R^2 is more relative to 1, the model performance is more robust. Therefore, the above evaluation indicators have the problem of inconsistency in tendency. The inconsistency of tendency refers to a trend that the stronger the model's prediction ability, the larger the evaluation index[17]. Therefore, this study transforms MAE, $MAPE$ and $RMSE$ as follows:

$$MAE' = \frac{1}{MAE} \tag{13}$$

$$MAPE' = \frac{1}{MAPE} \tag{14}$$

$$RMSE' = \frac{1}{RMSE} \tag{15}$$

However, the evaluation indicators of the above regression models have their pros and cons, resulting in the lack of comprehensive evaluation of different models by a single indicator, so a comprehensive evaluation indicator is needed. Zorlu et al.[18] proposed a comprehensive evaluation method based on multi-index ranking, which effectively overcomes the inconsistency of the tendency of traditional evaluation methods. However, this ranking-based method only considers the relative size relationship of the same evaluation index, ignoring the influence of the difference of the same evaluation index on the model evaluation. Therefore, this study proposes an improved comprehensive evaluation method based on the research of Zhang et al.[17] and Zorlu et al.[18]. The main steps are as follows:

Step1: Use the traditional regression model evaluation indicators to evaluate the model, calculate the difference Bias of each evaluation indicator on the training set and the test set, and calculate the mean of the differences and sort them from small to large;

Step2: Convert the indicators with inconsistency tendency to get MAE', $MAPE'$ and $RMSE'$;

Step3: Combine the values of MAE', $MAPE'$, $RMSE'$ and R^2 to rank each model on the training set and test set, respectively;

Step4: Sum up the rankings on the training set and test set, respectively and sort them from small to large;

Step5: Combine the rankings of Step1 and Step3 to get the comprehensive ranking of each model;

4. Results and discussion

4.1 Performance evaluation for weak learners

SVR and KNN models with initialization parameters are established based on the preprocessed data. Combined with the traditional regression model evaluation method to evaluate the model performance, the evaluation results shown in Figure 5 are obtained. For the SVR model, except for the $MAPE$ indicator, the other indicators are very close in the training and test sets. The difference between each indicator's training set and the test set is only 0.787 at most, so overall has a good generalization performance. At the same time, the $MAPE$ index of the SVR model on the test set is 24.664 percentage points, compared with 4.853 on the training set, a difference of 19.811 percentage points. However, this is insufficient to say that the SVR model suffers from significant bias or overfitting. Based on the definition of $MAPE$, it can be found that when $Y_{i_{\text{observed}}}$ is approached to zero, $MAPE$ will have a considerable value. This situation is related to the target variable (CL) minimum value of 0.029. matched. Therefore, this situation is caused more by the definition of $MAPE$ itself and the nature of the experimental data itself.

Correspondingly, for the KNN model, the values of its indicators on the training set and the test set are also very close, especially for the $MAPE$ index, which is only 7.169 on the test set; at the same time, the KNN model has a coefficient of determination on the test set. R^2 is 0.957, so it can be shown that the KNN model also has good prediction performance

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on the current data. Comparing the SVR and KNN models, it can be seen that on the current data, the SVR model has a strong generalization ability, and the deviation of the model between the training set and the test set is slight; while the KNN model is due to its "lazy" learning characteristics, the model The learning performance is relatively weak, but thanks to the advantages of the non-parameterized model, the KNN model is less affected by the features of the data set itself, so that it has a $MAPE$ index far lower than the SVR model; at the same time, except for R^2, the other indicators of the KNN model are lower than the SVR model.

Therefore, it can be seen from the model evaluation results that the SVR and KNN models are well suited for the current task scenario, and both models have their corresponding advantages. At the same time, the evaluation results also reflect the defect that a single evaluation index cannot comprehensively measure the model's performance. Therefore, this study further conducts model fusion and a comprehensive evaluation of model performance. But before that, we will perform optimization of model hyperparameters to ensure that weak learners participating in model fusion have relatively optimal hyperparameters.

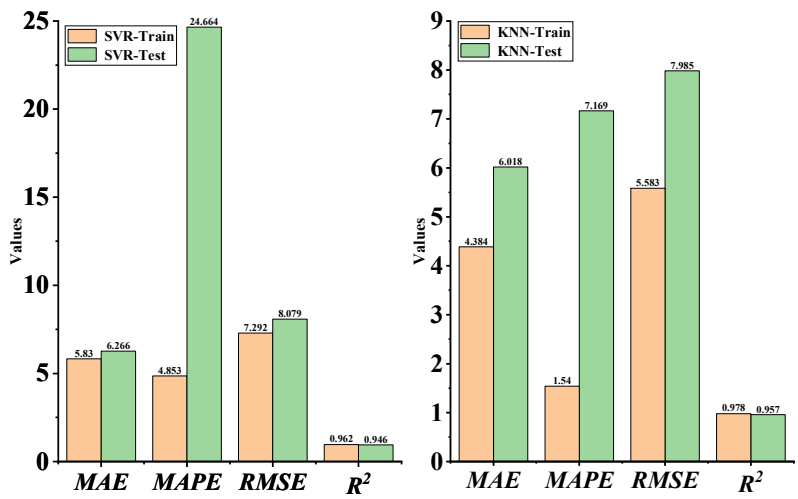

Figure 5 model evaluation

4.2 Hyperparameter optimization and training process

The Grid Search method will be very time-consuming when the model has many hyperparameters. Although the related Random Search has high efficiency, it is easy to lose the optimal value due to the randomness. Considering these reasons, this study used Bayesian optimization[19] to adjust the critical hyperparameters of SVR and KNN models. For the SVR model, the adjusted values of kernel, C(regularization factor) and Epsilon (relaxation variable) are Linear,1.2, and 0.3, respectively. For the KNN model, n_neighbors(the number of neighboring points), algorithm(used to calculate the adjacent points), P (the power exponent of Minkowski distance), and Leaf size are 4 ball tree, 2 and 25, respectively. Among them, the value of hyperparameter P of the KNN model is 2, indicating that KNN takes Euclidean distance as the metric index in the current task.

The fitting effect between the predicted and actual observed values of the SVR and KNN models on the test set after hyperparameter optimization is shown in Figure 6. Show the model performance has been improved to a certain extent. For the SVR model, the coefficient of determination R^2 on the test set has been increased by 1.2%. For the KNN model, the coefficient of determination R^2 on the test set is increased by 2.9%.

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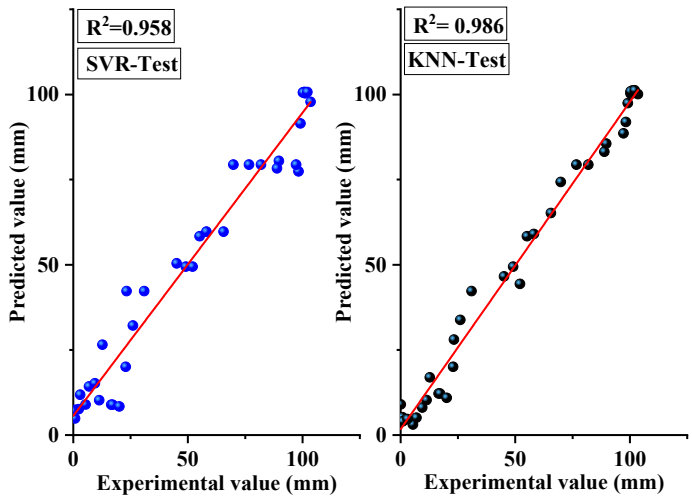

Figure 6 linear fitting of the model prediction results

4.3 Model fusion and Comprehensive evaluation

To overcome the lack of reliability in the prediction results of a single model and simultaneously realize the deep integration of the model decision-making layer, the weighted average method is used to fuse the adjusted SVR and KNN models. Therefore, the output result of the fusion model $H(x)$ is the weighted average of the predictions from the KNN and SVR model. Among them, it can be seen from Figure 5 and Figure 6 that the overall error of the KNN model is lower than that of the SVR model. After the hyperparameter adjustment, the test set determination coefficient R^2 of the KNN model is 0.028 higher than that of the SVR model. Therefore, this study assigns more weight to the KNN model.

Since MAE can well measure the deviation between the observed value and the true value, and has lower computational complexity than $RMSE$ [16], combined with the definition of Equation 8, 9, we can get $Y_{\text{predicted}} = H(x) = \frac{1}{T}\sum_{i=1}^{T} w_i h_i(x)$ So, The loss function of $H(x)$ is shown in Equation 16.

$$\min\left(\text{Loss}=\frac{1}{S}\sum_{i=1}^{S}\left|Y_{i_{\text{observed}}} - Y_{i_{\text{predicted}}}\right|\right) \text{s.t.} w_1 + w_2 = 1 \,\&\, w_1 \geq 0, w_2 \geq 0 \tag{16}$$

where $Y_{i_{\text{predicted}}}$ is the predicted value of the i_{th} sample by the fusion model $H(x)$. Through using the scientific computing tool Scipy[20] to solve the constrained minimum optimization problem corresponding to formula 16, the weight $w_1 = 0.183$ of the base model SVR and the weight $w_2 = 0.817$ of the KNN model is finally obtained.

Based on the model evaluation indicators in Section 3.4, the ensemble model $H(x)$ fitting results on the training set and the test set are shown in Figure 7. It can be seen from the figure that the coefficient of determination R^2 of the ensemble model $H(x)$ on the training set is 0.999, and the coefficient of determination R^2 on the test set is 0.997, compared with the R^2 of the SVR and KNN models on the test set, which increased by 0.041 and 0.011, respectively.

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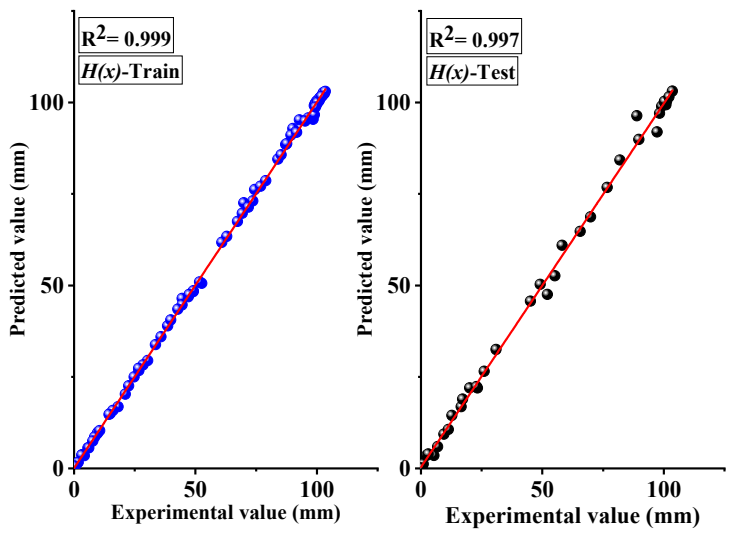

Figure 7 linear fitting of the model prediction results

Table 3 shows the detailed evaluation results of the integrated model $H(x)$. The average absolute error $MAPE$ of $H(x)$ on the test set is 0.149, which is 24.515 percentage points lower than that of the SVR model; compared with the KNN model, it is 7.02% lower. Percentage points indicate that the more significant weight of KNN makes it play an excellent lifting effect in the ensemble model. The MAE, $RMSE$, and other indicators have also decreased significantly. Therefore, the model evaluation results show that $H(x)$ has more reliable predictions than weak learners SVR and KNN.

Table 3 model evaluation metrics

Model	**Performance Metrics on Train Data**				**Performance Metrics on Test Data**			
	MAE	$MAPE$	$RMSE$	R^2	MAE	$MAPE$	$RMSE$	R^2
SVR	5.830	4.853	7.292	0.962	6.266	24.664	8.079	0.946
KNN	4.384	1.540	5.583	0.978	6.018	7.169	7.985	0.957
$H(x)$	0.134	0.037	0.198	0.999	1.361	0.149	2.080	0.997

Further, Table 4 is obtained from the statistical results in Table 3, where the symbol B represents the bias between the training set and the test set of each indicator, and the bias is used to reflect the model's generalization performance. It can be seen from the table that the $MAPE$ -B and R^2 -B of the ensemble model $H(x)$ are much smaller than the SVR and KNN models; at the same time, the MAE -B and $RMSE$ -B of the ensemble model $H(x)$ are more significant than the SVR model. Therefore, combined with the mean of the bias, the overall generalization performance of each model is ranked as $H(x)$>KNN>SVR;

Table 4 bias between model evaluation metrics

Model	**Bias of Performance Metrics**				
	MAE -B	$MAPE$ -B	$RMSE$ -B	R^2 -B	Average
SVR	0.436	19.811	0.787	0.016	5.263
KNN	1.634	5.629	2.402	0.021	2.422
$H(x)$	1.227	0.112	1.882	0.002	0.806

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After performing tendency consistency conversion for each evaluation index, the evaluation results of SVR, KNN, and $H(x)$ on the training and test sets are obtained, respectively, and the ranking of each index is shown in Figure. 8. On the training set, according to the relative size relationship between MAE', $MAPE'$, $RMSE'$ and R^2, the performance ranking of each model is $H(x)$ >KNN>SVR. And the same ranking result is also obtained on the test set; The comprehensive ranking results of the model on the training set and test set are consistent with Table 4. Therefore, in this study, the ensemble learning model $H(x)$ has the overall best performance index.

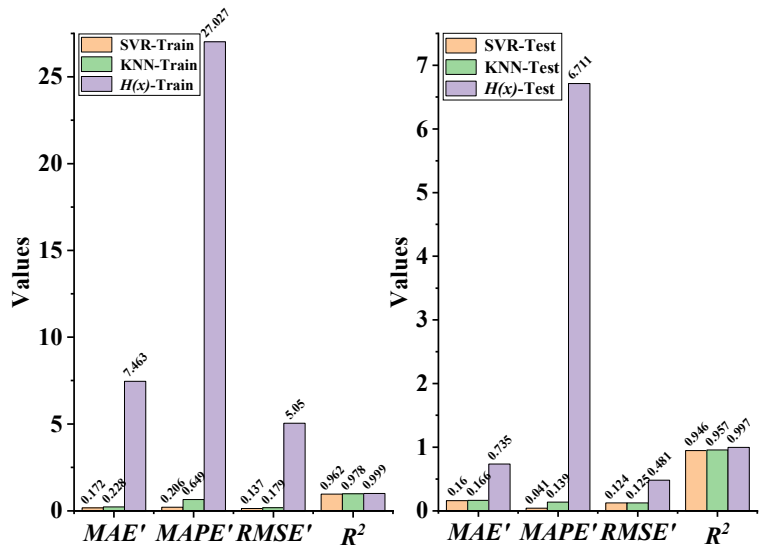

Figure 8 ranking of model performance

5. Conclusion

In this study, combined with the previous finite element analysis results, an experimental data set containing 125 samples were collected and constructed. After exploratory data analysis and preprocessing, the SVR and KNN models with initialization parameters were built to perform the tower Pre-bias prediction, combined with 5-fold cross-validation and Bayesian optimization, optimizes the hyperparameters of the model. At the same time, to further improve the reliability of the model prediction results, the model fusion strategy of the weighted average method is used to fuse the SVR and KNN models at the decision level, and the reliability of the model prediction results is improved through group decision-making. Among them, in the model fusion process, the ensemble model's loss function is defined in combination with MAE, and the fusion weight of the two base models is obtained by solving the constrained minimum optimization problem. Finally, each model is evaluated using the regression model evaluation index and a ranking-based comprehensive evaluation index. The evaluation results show that the model fusion dramatically improves the prediction performance of the base model and has a more robust generalization ability. Some of the contributions of this paper are as follows: (1) There is a trade-off between small sample size and model accuracy. While ensuring the accuracy of the model, the demand for data and computing power is reduced; (2) A comprehensive evaluation metric is improved and proposed, which achieves practical evaluation of each model while overcoming the problem of inconsistency in tendencies；

Overall, this research results can enhance and extend the traditional finite element analysis method, promoting the high-quality development of construction monitoring work by improving the prediction accuracy and efficiency. At the same time, the feasibility of modeling predictions on tasks with small data volumes is verified. However, the auxiliary decision-making role of machine learning in construction control and industrial process monitoring is still at a low level. Although some existing studies have achieved good accuracy, how to balance the model accuracy with the safety control in the field, and achieve closed-loop control between model prediction and industrial practice, still needs further in-depth research. In future practical applications, various on-site influencing factors and expert evaluation experience will be fully considered to design more representative and broader experimental datasets and more valuable and feasible comprehensive model performance evaluation indicators. Thereby promoting the more in-depth and practical application of machine learning in the engineering field.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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